G12Adv	Mathematics	الرياضيات	T1 - 2023/24

Multiple Choice Part	جزء الاختيار من متعدد
Show all your work when answering these questions.	يجب كتابة خطوات الحل التفصيلية للمفردات الاختبارية كافة.

Multiple Choice — اختيار من متعدد

1. Determine the interval(s) where $f(x) = \dfrac{\sqrt{x-1} + e^x}{x^2 - 2}$ is continuous.

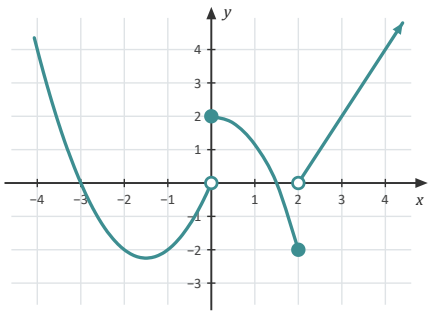
A) $(-1, \sqrt{2}) \cup (\sqrt{2}, \infty)$

B) $[-1, \sqrt{2}) \cup (\sqrt{2}, \infty)$

C) $[1, \sqrt{2}) \cup (\sqrt{2}, \infty)$

D) $(1, \sqrt{2}) \cup (\sqrt{2}, \infty)$

2. Use the graph to determine $\lim\limits_{x \to 2^+} f(x)$, if it exists.



A) -2

B) -1

C) 0

D) 2

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Multiple Choice — اختيار من متعدد

3. For $f(x)=\begin{cases} ax+b, & x<0 \\ x^2-3x, & x\geq 0 \end{cases}$ find all real numbers a and b such that $f'(0)$ exists.

A) $a=-3,\ b=0$

B) $a=0,\ b=-3$

C) $a=0,\ b=3$

D) $a=3,\ b=0$

4. Find the derivative of the function $f(x)=4\sec^{-1}(x^4)$.

A) $f'(x)=\dfrac{16}{x\sqrt{x^8-1}}$

B) $f'(x)=\dfrac{4}{x\sqrt{x^8-1}}$

C) $f'(x)=\dfrac{16}{x\sqrt{x^4-1}}$

D) $f'(x)=\dfrac{4}{x\sqrt{x^4-1}}$

5. Find all vertical and slant asymptotes of $f(x)=\dfrac{x^3}{x^2+x-6}$.

A) $x=-2,\ x=3,\ y=x-1$

B) $x=-3,\ x=2,\ y=x+1$

C) $x=-2,\ x=3,\ y=x+1$

D) $x=-3,\ x=2,\ y=x-1$

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Multiple Choice	اختيار من متعدد

6. **Evaluate**

$$\lim_{x\to 3} \frac{x^3 - 27}{x^2 - 9}$$

A) $\frac{3}{2}$

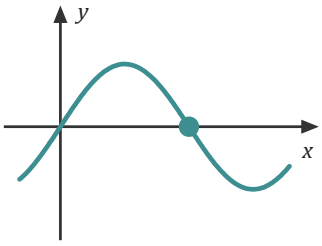
B) 2

C) 6

D) $\frac{9}{2}$

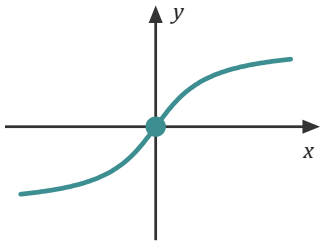
7. **Which graph does not have a tangent line at the given point?**

A)



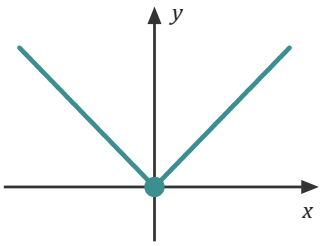
y = sin x at x = π

B)



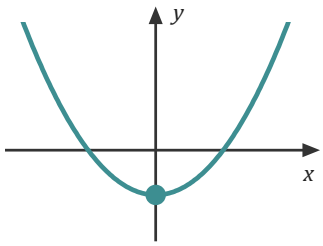
y = $\tan^{-1}$x at x = 0

C)



y = |x| at x = 0

D)



y = x² − 1 at x = 0

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Multiple Choice	اختيار من متعدد

8. Use logarithmic differentiation to find the derivative of $f(x) = x^{(4-x^2)}$.

A) $f'(x) = x^{(4-x^2)}\left[-2x\ln x + \frac{1}{x}(4-x^2)\right]$

B) $f'(x) = x^{(4-x^2)}\left[-2x\ln x + 4 - x^2\right]$

C) $f'(x) = x^{(4-x^2)}\left[-\ln x + \frac{1}{x}(4-x^2)\right]$

D) $f'(x) = -2x\ln x + \frac{1}{x}(4-x^2)$

9. Find all values of x for which the tangent line to the curve $y = 3x^4 - 6x^2 + 1$ is horizontal.

A) $x = -\sqrt{2},\ 0,\ \sqrt{2}$

B) $x = -\sqrt{3},\ 0,\ \sqrt{3}$

C) $x = -2,\ 0,\ 2$

D) $x = -1,\ 0,\ 1$

10. The function $s(t) = 3t^3 + t$ represents the position in feet of an object at time t seconds. Find the average velocity between $t = 0$ and $t = 2$.

A) $v_{avg} = 24$ ft/s

B) $v_{avg} = 26$ ft/s

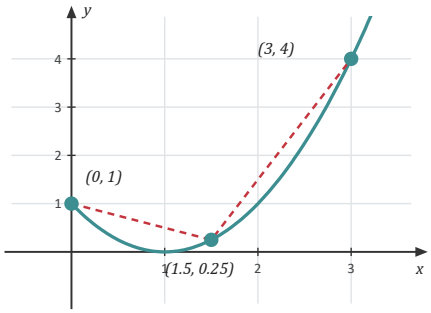
C) $v_{avg} = 13$ ft/s

D) $v_{avg} = 12$ ft/s

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Multiple Choice	اختيار من متعدد

11. **Estimate the arc length of the curve $y = (x-1)^2$ on the interval $0 \le x \le 3$ using $n = 2$ line segments.**



A) 2.8125 **B)** 4.0359

C) 5.7159 **D)** 32.6717

12. **The function $s(t) = 2\sqrt{t} + 2t^2$ represents the position of an object. Find the acceleration function.**

A) $a(t) = \frac{1}{2}t^{3/2} + 4$

B) $a(t) = -\frac{1}{2}t^{-3/2} + 4$

C) $a(t) = t^{-1/2} + 4t$

D) $a(t) = t^{1/2} + 4t$

13. **Find the derivative of the function $f(x) = \sin^2(\tan x)$.**

A) $f'(x) = (\sec^2 x)[2\sin(\tan x)][\cos(\tan x)]$

B) $f'(x) = (\sec^2 x)[\sin(\tan x)][\cos(\tan x)]$

C) $f'(x) = (\sec x)[2\sin(\tan x)][\cos(\tan x)]$

D) $f'(x) = (\tan x \sec x)[2\sin(\tan x)][\cos(\tan x)]$

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14. **Given that the function $h(x) = \dfrac{x^2}{f(x)}$, if $f(2) = 2$ and $f'(2) = 1$ compute $h'(2)$.**

A) $h'(2) = -5$ **B)** $h'(2) = -1$

C) $h'(2) = 1$ **D)** $h'(2) = 3$

15. **Given that the function $f(x) = x^3 + 2x + 1$ has an inverse function g, compute $g'(-2)$.**

A) $g'(-2) = -1$ **B)** $g'(-2) = 5$

C) $g'(-2) = \frac{1}{5}$ **D)** $g'(-2) = 0$

16. **The concentration of a certain chemical after t seconds of an autocatalytic reaction is given by $c(t) = \dfrac{6}{2e^{-8t} + 1}$. (a) Show that $c'(t) > 0$. (b) Use this information to determine that the concentration of the chemical never exceeds 6.**

A) c'(t) > 0 and the concentration of the chemical never exceeds 6

B) c'(t) < 0 and the concentration of the chemical never exceeds 6

C) c'(t) > 0 and the concentration of the chemical exceeds 6

D) c'(t) > 0 and the concentration of the chemical never exceeds 3

Paper Part	الجزء الورقي
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Question	F1a	السؤال

Suppose that $f(x)$ is bounded: that is, there exists a constant M such that $|f(x)| \leq M$ for all x. Use the Squeeze Theorem to prove that $\lim_{x \to 0} x^2 f(x) = 0$.

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Question	F1b	السؤال

Evaluate

$$\lim_{x\to\infty}\left(\sqrt{4x^2-2x+1}-2x\right)$$

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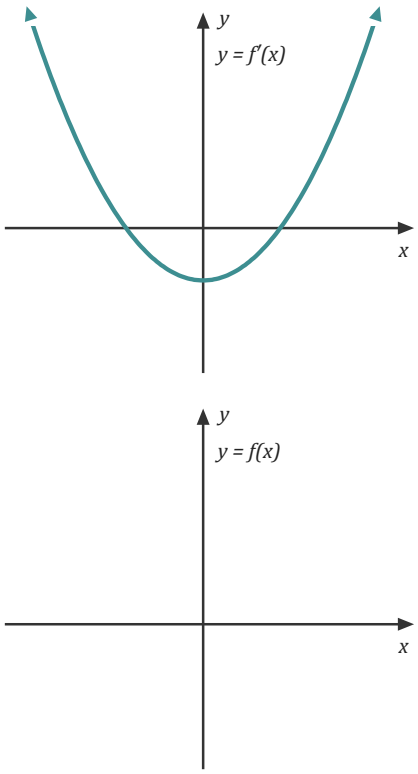
Question	F2a	السؤال

Find the derivative of $f(x) = \dfrac{3}{x+1}$ at an unspecified value of x using $f'(x) = \lim\limits_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$.

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Question	F2b	السؤال

Use the given graph of $f'(x)$ to sketch a plausible graph of the continuous function $f(x)$.



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Question	F4	السؤال

Find an equation of the tangent line to $y^2 + xe^y = 4 - x$ at the point $(2, 0)$.

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Question	F5	السؤال

Find a value of c as guaranteed by the Mean Value Theorem for the function $f(x) = x^3 + x$ on the interval $[-1, 0]$.

End of Questions **انتهت الأسئلة**